

JUAN DIEGO MENDEZ

+1(786) 824-8479 ◊ Austin, TX

juandiegomendezgutierrez@gmail.com ◊ linkedin.com/in/juan-diego-mendez ◊ jdmg1308.github.io/site

EDUCATION

Bachelor of Computer Science, The University of Texas at Austin 2022 - 2025

Relevant Coursework: Cyber-Physical Systems, Programming for Performance, Computer Vision, Principles of Computer Systems (OS), Computer Architecture, Intro to Probability and Statistics

SKILLS

Programming Languages	C#, Python, C, C++, JavaScript, TypeScript, Java, HTML, CSS
Soft Skills	Problem Solving, Resourcefulness, Adaptability, Mission-Driven, Goal-Oriented
Tools	Docker, Git, WebSockets, Bitbucket, VS Code, CI/CD, WSL, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Linux, Jira

EXPERIENCE

Software Engineer May 2025 - Present
Radical Empathy Education Foundation *Austin, TX*

- Modernized and refactored a Unity-based VR abuse prevention training experience (“TRAPPED”), used by over 15,000 students and professionals. Migrated the project to the latest Unity version, integrated Meta XR and XR Plugin systems, and resolved compatibility issues to support modern headset SDKs.
- Continuing development to enhance interaction fidelity and long-term technical stability for REEF’s VR-driven training platform.

Team Lead for VR Design Sep 2024 - May 2025
Longhorn Neurotech *Austin, TX*

- Led the development of an interactive VR environment using Unity 3D, C#, and the Meta XR SDK, designed to interface in real time with EEG-based brain-computer input.
- Designed and implemented a Unity input controller capable of handling up to four—and potentially six—degrees of EEG-based input, tailored for flexible and immersive neuro-interaction.
- Collaborated with signal processing and ML teams to route preprocessed brainwave data from a Python-based pipeline into Unity via WebSocket networking, enabling low-latency, real-time control within the VR system

Software Engineer Jun 2023 - Aug 2023
Monkeyflux S.L. *Madrid, Spain (remote)*

- Improved the frontend experience of the MonkeyFlux Babuin platform using TypeScript, HTML, and CSS by building reusable components, resolving UI bugs, and optimizing performance.
- Developed and deployed a Google Forms integration using Google’s API to automate bug reporting into the company’s internal tracking system, streamlining DevOps workflows.
- Collaborated with the DevOps team on environment setup using Docker, Node.js and WSL, participated in code reviews, and contributed to backlog grooming in Jira and Bitbucket.

PROJECTS

XR Environment Interactions using a Depth Camera. Built a real-time XR system enabling physical interaction with virtual objects. Integrated an Oculus Quest 2 and an Intel RealSense D435 for hardware-based depth sensing and spatial awareness, and applied voxel-based optimization techniques to boost performance. Successfully documented the project in an [academic paper](#) as part of a research course.

EEG-Based Reinforcement Learning for Binaural Beat Optimization. Built a pipeline using Python and C++ for low-latency, real-time EEG Processing. Applied reinforcement learning to optimize binaural beat frequency and phase shifts on an audio based on EEG responses. Integrated Muse S Athena headset using OSC via the Muse SDK to stream data, achieving a 30% increase in meditation metrics during personal testing.